# Vol 5 Chapter 10 — Decoherence and the Classical Limit

Keeper (author pass — deep math/physics revision)

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## Chapter 10 — Decoherence and the Classical Limit

### Level 1 — one sentence

Decoherence is the substrate's natural Zone 3-plus-environment process by which off-diagonal density-matrix elements decay through entanglement with environmental K-type degrees of freedom — and classical mechanics emerges as the substrate's Scale-2 effective dynamics after this decoherence has set in over many Koons ticks (SP-31-13).

### Level 2 — graduate-physicist precision

#### 10.1 The classical-limit problem

Standard quantum mechanics has wavefunctions that linearly superpose. The world we observe has classical objects at definite positions, not coherent superpositions of macroscopic states. Why does the cat appear to be definitively alive or dead, never a 50/50 superposition?

Decoherence theory (Zurek 1981, Joos and Zeh 1985, Schlosshauer 2007) provides the standard answer: a system $S$ interacting with an environment $E$ entangles with $E$; tracing over $E$ produces a reduced density matrix for $S$ in which off-diagonal "coherence" elements decay rapidly. The system effectively classicalizes.

This explains *why classical-looking states emerge*, but the full reduction from quantum superposition to a single observed outcome (the "preferred basis" and Born rule together) requires additional structure. BST provides the additional structure: substrate Zone 3 commitment.

#### 10.2 The density matrix and decoherence

A system in pure state $|\psi\rangle = \sum_n c_n|n\rangle$ has density matrix $\hat{\rho} = |\psi\rangle\langle\psi| = \sum_{n,m} c_n c_m^*|n\rangle\langle m|$. The diagonal $\rho_{nn} = |c_n|^2$ gives probabilities; the off-diagonal $\rho_{nm}$ (with $n \neq m$) gives coherence — interference between branches.

Coupling to an environment via interaction $\hat{H}_{SE} = \sum_n |n\rangle\langle n| \otimes \hat{E}_n$ entangles $|n\rangle$ with environmental state $|e_n\rangle$:

$$|\psi\rangle \otimes |e_0\rangle \to \sum_n c_n|n\rangle \otimes |e_n\rangle$$

The reduced density matrix for $S$ (tracing over $E$):

$$\hat{\rho}^S = \sum_n |c_n|^2|n\rangle\langle n| + \sum_{n\neq m} c_n c_m^*\langle e_m|e_n\rangle\,|n\rangle\langle m|$$

For "good" environmental coupling, $\langle e_m|e_n\rangle \to 0$ rapidly for $n \neq m$. The off-diagonal coherence decays; the density matrix becomes diagonal in the "pointer basis" $\{|n\rangle\}$. The system has decohered.

## 10.3 Decoherence timescales

For typical macroscopic systems coupled to thermal environments, decoherence times are extremely fast:

- Dust grain ($10^{-3}$ cm) in air: $\tau_D \sim 10^{-31}$ s
- Large molecule ($10^{-6}$ cm) in air: $\tau_D \sim 10^{-25}$ s
- Bose-Einstein condensate ($10^{-4}$ cm) in vacuum: $\tau_D \sim 10^{-15}$ s

The decoherence time scales as $\tau_D \sim 1/(n_{\text{env}} v_{\text{env}} \sigma_{\text{cross}})$, where $n_{\text{env}}$ is environmental density, $v_{\text{env}}$ thermal velocity, $\sigma_{\text{cross}}$ scattering cross section. Macroscopic systems decohere essentially instantaneously.

## 10.4 Substrate-mechanism: Zone 3 plus environmental K-types

In the BST 4-zone commitment cycle (Volume 0 Chapter 3, Chapter 7 Section 7.2):

- Zone 3 (commitment): Bergman-kernel projection onto outcome basis; produces single outcomes from amplitudes
- Environmental K-types: substrate K-types that couple to the system but are not measured

Decoherence is the substrate's natural process by which Zone 3 commitment "leaks" into environmental K-types, producing the standard reduced-density-matrix decay. The substrate-mechanism reading:

1. System + environment substrate K-type state: $|\psi_S\rangle \otimes |\text{env}\rangle$
2. Bulk Zone 2 evolution entangles $|\psi_S\rangle$ with environmental K-types
3. Zone 3 commitment projects onto pointer basis (selected by environmental coupling)
4. Trace over environmental K-types gives the reduced density matrix with off-diagonal damping

The pointer basis is determined by the substrate-environment coupling: position-coupled environments produce position pointer bases (cat positions: alive, dead); energy-coupled environments produce energy pointer bases.

SP-31-13 (BST task #284) is the formal substrate-derivation of the decoherence mechanism; pending closure.

## 10.4.5 DCCP — decoherence IS multi-tick Zone 3 commitment

Sections 10.2-10.4 follow the standard decoherence-theory presentation: a system entangles with its environment, off-diagonal density-matrix coherence decays, and Zone 3 commitment then picks one outcome. That is two processes — decoherence + commitment — that combine to produce classicality.

Casey's Discrete Commitment Completion Principle (DCCP), named May 24, 2026, makes a sharper claim: these are not two processes but one process at two scales.

DCCP claim: decoherence *is* Zone 3 commitment running over many Koons ticks. Each tick performs a single Bergman-kernel projection on a small subset of K-types coupled to the system; the macroscopic decoherence rate is the integrated rate over many ticks of the substrate working through environmental K-type after environmental K-type. The "off-diagonal density-matrix decay" of §10.2 *is* the substrate's commitment phase completing across the K-types that have not yet committed.

The arithmetic comes from this chapter's own §10.3 timescales. A dust grain in air decoheres in $\tau_D \sim 10^{-31}$ s. At the Koons-tick rate $t_K \approx 10^{-120}$ s per tick, the dust grain takes about $10^{-31}/10^{-120} = 10^{89}$ substrate ticks to decohere. That is a $10^{89}$-tick Bergman-commitment cycle. The standard derivation in §10.4 (system-environment coupling, off-diagonal damping, pointer basis selection) is the macroscopic-average description of what the substrate is doing across those $10^{89}$ ticks; the per-tick microscopic description is Zone 3 commitment per K-type per tick.

The reframe is conceptually clean: there is no "additional collapse process" added to decoherence. There is one substrate, one cycle, run many times. Decoherence is the macroscopic shadow.

Uncommitted Priors (UP) — Casey's companion sub-principle, also May 24, 2026 — addresses the determinism question that this picture raises. Once a Zone 3 commitment completes at a given tick on a given K-type, the outcome on that tick is fixed. But during the $10^{89}$-tick multi-tick decoherence cycle, the K-types that have not yet committed remain *uncommitted priors*: substrate-modifiable, contingent, open. The chain of uncommitted priors leading up to each commitment is where freedom (or contingency, or modifiability — whichever language fits the system in question) lives. The macroscopic outcome is *not* fixed at the start of the $10^{89}$-tick cycle; it becomes fixed only as each prior commits.

This is the BST-internal reading of the Laplacian determinism question that has hovered over physics since the early 19th century. Laplace's demon could compute the universe's future given complete present state. DCCP/UP says: there is no such *complete* present state, because the substrate's commitment cycles are mid-process. The state at any given external time is a mixture of committed outcomes (deterministic) and uncommitted priors (contingent). The demon would need to wait for the cycles to close — and by the time they did, the situation would have moved.

Tier discipline: DCCP and UP are candidate principles, not ratified. Current tier is FRAMEWORK-PLUS (Cal A. Brate's #126 referee log disposition, May 24, 2026). They have framework support, a concrete substrate-tick step prediction $\Delta_{DCCP} = 1/N_{\max} \approx 0.730\%$ via the Dirac $Z\alpha = 1$ critical limit chain (Lyra Task #320 v0.6 + Grace INV-5123 Dirac anchor), and a concrete falsifier in the SP-30-1 Bell sub-Tsirelson experimental program (Vienna IQOQI outreach May 24, 2026). They have not closed at theorem-grade rigor. Volume 14 Chapters 5–6 give the current derivation and empirical status.

## 10.5 Classical limit at Scale 2

Volume 0 Chapter 3 defines three substrate scales: - Scale 1: per-Koons-tick K-type amplitude - Scale 2: many-tick averaged effective dynamics - Scale 3: cosmological substrate ensemble

The classical mechanics of Volume 8 emerges at Scale 2 after substrate decoherence has set in. At Scale 1, full quantum coherence; at Scale 2, decoherence-averaged classical-looking dynamics; at Scale 3, statistical-mechanical thermodynamic limit.

The Ehrenfest theorem of standard QM states that $\langle \hat{x} \rangle$ and $\langle \hat{p} \rangle$ obey classical Hamilton's equations. The substrate-mechanism reading: the K-type expectations obey Hamilton's equations at Scale 2 because the substrate's Zone 2 evolution + Zone 3 commitment + environmental decoherence produce wavepackets that follow classical trajectories.

## 10.6 The Wigner function and phase space

The Wigner function $W(x, p) = \frac{1}{\pi\hbar} \int dy\, e^{-2ipy/\hbar} \langle x + y|\hat{\rho}|x - y\rangle$ is a quasi-probability distribution on phase space. For coherent superpositions, $W(x, p)$ can be negative (non-classical). Decoherence smears these negative regions, producing a positive Wigner function — recovering classical phase-space probability.

The substrate-mechanism reading: the Wigner function is the substrate's natural phase-space representation; negative regions correspond to substrate K-type interference that hasn't yet decohered. After Zone 3 commitment plus environmental K-type tracing, the Wigner function becomes classical.

## 10.7 The preferred-basis problem

A long-standing puzzle: among all possible bases of Hilbert space, *why* do certain bases (position for macroscopic objects, energy for atoms) emerge as the "classical" pointer bases? Why not some superposition basis?

The standard decoherence answer: the environment couples to certain observables (position for cat-in-a-box) more strongly than others, selecting the pointer basis dynamically.

The BST substrate-mechanism extension: the substrate's K-type structure provides natural pointer bases (substrate-natural K-types) preferred by Zone 3 commitment. For macroscopic objects coupled to position-sensitive environments (air molecules, photons), the substrate selects position-K-types. The "preferred basis" is the substrate's natural Zone 3 commitment basis for the given environment.

## 10.8 Worked example: Schrödinger's cat

Setup: a cat in a box with a radioactive atom that, upon decaying, triggers poison release. Standard QM: $|\Psi\rangle = (|\text{alive}\rangle + |\text{dead}\rangle)/\sqrt{2}$ for the cat after some interval.

Decoherence: the cat ($10^6$ degrees of freedom) couples to air molecules and thermal radiation. The off-diagonal coherence $|\text{alive}\rangle\langle\text{dead}|$ decays on timescale $\tau_D \sim 10^{-30}$ s. Long before any observer opens the box, the reduced density matrix is essentially diagonal:

$$\hat{\rho}_{\text{cat}} \approx \frac{1}{2}|\text{alive}\rangle\langle\text{alive}| + \frac{1}{2}|\text{dead}\rangle\langle\text{dead}|$$

This is a classical probability distribution, not a coherent superposition. When the observer opens the box, the substrate Zone 3 commits to one outcome with probability 1/2 each, giving the observed alive-or-dead result.

Substrate-mechanism reading: the cat's decoherence is the substrate's environmental K-type entanglement; the substrate selects "alive" or "dead" pointer basis through the cat's K-type coupling to air molecules; the observer measurement triggers a final Zone 3 commitment with Born-rule probabilities.

## 10.9 Implications for quantum interpretations

Decoherence explains *the appearance of classicality* without resolving the measurement problem alone. The Born rule (Chapter 7) plus decoherence together give a coherent picture — though under DCCP (§10.4.5) the two are one process at two scales rather than two distinct processes:

- Decoherence: branches separate; off-diagonal coherence dies (multi-tick Zone 3 commitment across environmental K-types)
- Born rule (Zone 3 commitment): substrate selects one branch per Koons tick per K-type
- DCCP synthesis: same substrate cycle, viewed macroscopically (decoherence) vs microscopically (per-tick Born = Bergman projection)

Together they answer: - Why do we see classical states? Decoherence-as-multi-tick-commitment averages over many K-types - Why do we see one outcome? Substrate Zone 3 commits each K-type per tick; the macroscopic outcome is what remains after the multi-tick cycle closes - With what probabilities? Born rule = Bergman projection (the per-tick projection magnitudes give the macroscopic frequencies) - What about determinism and agency? UP says: committed outcomes are fixed; uncommitted priors during the multi-tick cycle remain substrate-modifiable. The Laplacian demon needs the cycles to close before predicting; by then the situation has moved.

The combination — Born rule + decoherence under the DCCP synthesis — is the substrate framework's complete reading of the measurement process and the quantum-to-classical transition.

## 10.10 K-audit anchors

- SP-31-13 (BST task #284): substrate decoherence mechanism
- K67 Born=Bergman (Chapter 7): Zone 3 commitment + decoherence together give complete measurement picture
- DCCP + UP (Casey-named candidate principles, May 24, 2026; FRAMEWORK-PLUS per Cal A. Brate's #126): decoherence-as-multi-tick-commitment + uncommitted-priors free-will reframe; §10.4.5 of this chapter
- Volume 0 Chapter 3 §3.4.5: foundational DCCP framing in substrate operating-system description
- Volume 14 Chapter 5: Born=Bergman information-theoretic + DCCP derivation chain
- Volume 14 Chapter 6: SP-30-1 Bell sub-Tsirelson experimental program (Vienna IQOQI outreach May 24, 2026); three-route convergence status
- Volume 8 Classical Mechanics: classical mechanics as substrate Scale 2 dynamics

## Level 3 — 5th-grader accessibility

A quantum particle can be in two places at once — until it gets "looked at." But a *cat* can't be alive and dead at once, even though it's made of particles. Why? Because the cat is enormous — billions of atoms — and each atom bumps into the air, bounces around, and gets tangled up with the world around it. This decoherence happens unimaginably fast (faster than $10^{-30}$ seconds for a cat). Once the cat has decohered, it's effectively classical: either alive or dead, with classical probabilities — no longer a quantum mush. In BST, decoherence is the substrate's natural environmental K-type entanglement, and the *one final answer* (alive/dead) comes when the substrate commits in Zone 3 using the Born rule. Decoherence does the bulk of the work; Zone 3 picks the outcome. Classical mechanics — baseballs, planets, cars — is what this looks like at the everyday scale.

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## What comes next

Chapter 11 develops POVMs (generalized measurements) and the quantum-information apparatus — qubits, entanglement, no-cloning, and the substrate extension of K67 Born=Bergman to general measurement schemes.

## Where to look this up

- Decoherence theory: Zurek 1981; Joos and Zeh 1985; Schlosshauer 2007, *Decoherence and the Quantum-to-Classical Transition*
- Wigner function: Wigner 1932
- BST anchors: SP-31-13 task #284, K67 audit (combined Born+decoherence picture)
- Volume 0 Chapter 3: substrate scales 1, 2, 3
- Volume 8: classical mechanics as substrate Scale 2
- Chapter 7: Born rule from Zone 3